

# Coursework Report Template for Module INM433 “Visual Analytics”

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**Abstract**—This study looked into how time and weather impact bike-sharing use. Finding patterns, examining non-linear correlations, and developing prediction models were the goals. To do this, a combination of computational techniques and visual analytics was done. Trends and relationships were discovered through the initial research study, and more information was collected by converting timestamps to features. To adjust for non-linear effects, polynomial regression was applied. A Random Forest model, adjusted with GridSearchCV, increased prediction accuracy and decreased the RMSE from more than 1000 to 348.23.

Temperature and humidity are important variables in the research. The data showed that up until a certain point, warmer temperatures encouraged bike use; after that, severe heat reduced physical activity. High humidity had a negative impact on usage. According to seasonal patterns, summer was the busiest time of year, while winter was the least active. The importance of linking data analysis and visualisation to support decision-making is shown by this project. Although the use of grid search was effective, its processing limitation and lack of geographical data caused issues. By increasing the number of data and advanced machine learning algorithms, future research could help improve urban transportation plans better.

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## 1 PROBLEM STATEMENT

Systems for sharing bikes have developed to be an essential component of urban mobility, providing a flexible and economical choice for transport. It's crucial to effective system planning, station management, and user satisfaction to understand the factors that impact bike usage. This study aims to analyse a dataset that includes public bike sharing to identify important trends and relationships between bike usage and different contextual, temporal, and weather variables (Zhang & Pan, 2018).

With a focus on analysing the public bike sharing, this research aims to answer the following questions:

- How can we predict bike-sharing counts by using weather and temporal features, and which hyperparameter optimisation methods are most effective in enhancing the accuracy of these predictions?
- What are the nonlinear effects of weather variables on bike-sharing counts, and how do these effects vary across different seasons and weather conditions?

The dataset used in this research contains timestamped records of bike-sharing counts and associated environmental and contextual variables, such as real temperature and supposed temperature, humidity, wind speed, category of the weather, holiday indicators, weekend indicators and seasons. These types of data such as temporal, climatic and categorical data are suitable for answering the research questions, as they capture the complex nature of factors influencing the bike-sharing behaviour.

The study applies visual analytics to investigate usage trends, weather effects, and prediction models to enhance bike-sharing operations and urban mobility techniques while providing useful and significant insights for system management.

## 2 STATE OF THE ART

Polynomial regression is an effective tool for evaluating non-linear relationships between variables, making it ideal for working out how time and weather affect bike-sharing use. Using features like variable relations or squared terms, this approach provides an understanding of how factors like humidity and temperature impact demand (Shashank, et al., 2023). For example, despite high temperatures that may increase use, polynomial regression shows this pattern. This approach improves machine learning by providing a greater understanding of the methods that temporal and environmental factors affect bike-sharing behaviour.

Several studies have examined the impact of weather on bike use, and techniques such as Random Forest have shown interest in evaluating weather-related predictions. It has been shown that depending on the time of day and season, factors including temperature, humidity, and wind speed have different impacts on bike usage. For example, while lower temperatures may only have noticeable effects in the winter, higher humidity could reduce usage during hot weather. Random Forest models can increase prediction accuracy and support feature selection, which is important for optimising hyperparameters in forecasting models (Feng & Wang, 2017).

Another method is to predict bike-sharing demand using context-aware prediction models, e.g. Recurrent Neural Networks (RNNs), which include spatial, temporal and meteorological data. The relationship of multiple variables, such as during severe weather conditions or periods of high demand, is accurately predicted by using this method (Lee & Ku, 2022). By improving the understanding of these models using visual analytics, researchers can investigate how weather and time combine to affect bike usage. Visualisation and contextual data both combine to ensure results are clear and improve models.

Building on the results of recent studies, this project will improve our understanding and prediction of bike-sharing usage by using machine learning and visual analytics. This will be achieved by developing a prediction model using polynomial regression that performs well for modelling the non-linear relationships between temporal variables, weather, and bike-sharing counts. The model's hyperparameters will be changed using a grid search method, which will evaluate multiple parameter combinations to find the most effective and accurate setup.

This approach shows the value of combining computational and visual methods. A greater understanding of the variables affecting demand will be possible due to the machine learning model that has been improved using grid search. A comprehensive understanding of how factors such as temperature, humidity, and wind speed interact with time to influence bike-sharing activity can be used by utilising polynomial regression to capture the complex non-linear effects. This combination of techniques ensures that the results are not only valid but also clear and useful.

In conclusion, a solid foundation for evaluating and predicting bike-sharing trends is provided through using machine learning with visual analytics. This study investigates how meteorological and temporal variables influence bike-sharing behaviour and provides helpful ideas for improving system management and planning by changing approaches from prior research.

### 3 PROPERTIES OF THE DATA

The dataset used in this study includes a variety of temporal and environmental factors and is focused on bike-sharing activities. With 17,415 rows and 10 columns, it provides an in-depth view of bike-sharing use by recording daily patterns over several years. This dataset is taken from a publicly accessible resource that combines information from weather stations and bike-sharing systems. The fields and type for the data:

1. "timestamp" - timestamp field for grouping the data
2. "cnt" - the count of new bike shares
3. "t1" - real temperature in C
4. "t2" – the temperature in C "feels like"
5. "hum" - humidity in percentage
6. "wind\_speed" - wind speed in km/h
7. "weather\_code" - category of the weather. 1 = Clear; mostly clear but have some values with haze/fog/patches of fog/ fog in the vicinity 2 = scattered clouds / few clouds 3 = Broken clouds 4 = Cloudy 7 = Rain/ light Rain shower/ Light rain 10 = rain with thunderstorm 26 = snowfall 94 = Freezing Fog
8. "is\_holiday" - Boolean field - 1 holiday / 0 non-holiday
9. "is\_weekend" - Boolean field - 1 if the day is a weekend

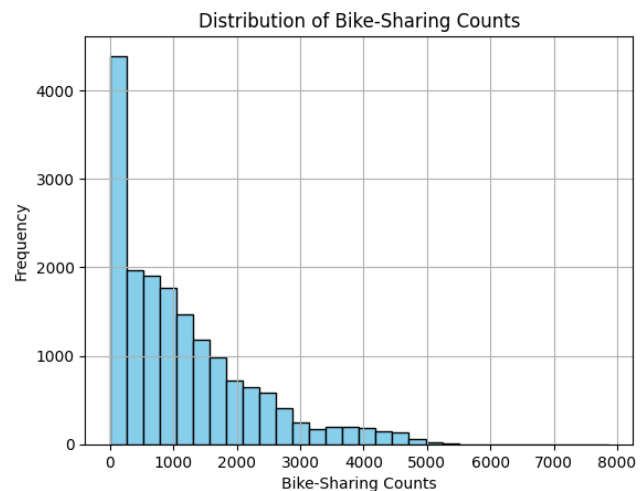
10. "season" - category field meteorological seasons: 0-spring; 1-summer; 2-fall; 3-winter.

The dataset is timed over multiple years, which offers information into long-term patterns. This dataset is relevant for bike-sharing systems in London. To analyse patterns, the data is gathered every day and every hour of the day.

The data looks like it has been collected from a combination of bike-sharing system logs, weather station reports and public holiday records. Therefore, some bias may exist, such as reduced activity during extreme weather or holidays, which could influence the trends in the data.

When reviewing the dataset, it was identified that the data has no missing values. For detecting outliers, a function was defined using the interquartile range and found that there were a few outliers, however, compared to the whole dataset, it didn't affect the values. The histogram in Figure 1 shows that the distribution of the data is rightly skewed, showing that the bike-sharing counts frequency of the lower values was the most common across the whole data. The heatmap in Figure 2 showed that there were no significant temporal gaps throughout the whole dataset, and they didn't have a major impact on the whole dataset.

This dataset provides a strong basis for solving the research questions. It allows us to explore trends, non-linear correlations, and prediction models by combining meteorological, temporal, and categorical data. However, to ensure the accuracy of the results, biases will be considered throughout the study, such as reduced activity during severe weather.



**Figure 1- Distribution of Bike-Sharing counts**

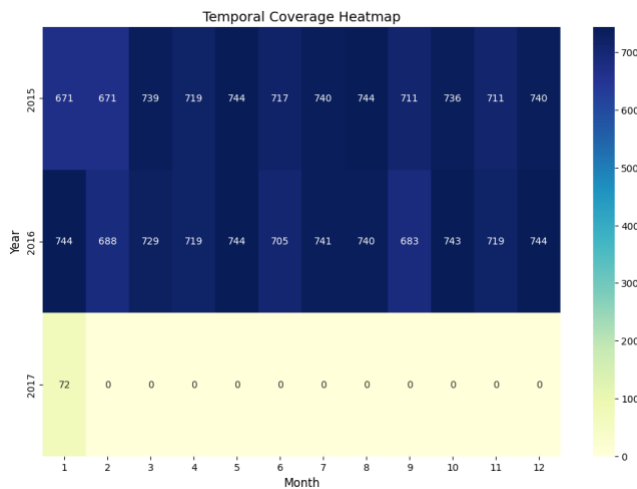


Figure 2- Temporal Coverage Heatmap

## 4 ANALYSIS

### 4.1 Approach

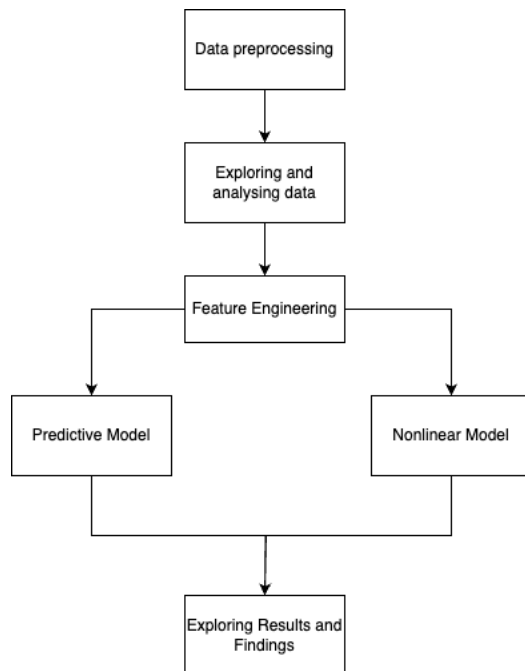


Figure 3- Workflow diagram

#### 1. Data Preprocessing:

- **Approach:** A lot of these data need some preprocessing to solve the problem of this project. The dataset has 10 columns and 17415 rows.
- **Computer operation:** Check for NaN values and if the dataset is complete and convert the timestamps column to datetime objects. Use Pandas library for data manipulation and preprocessing.
- **Human Analyst:** Understanding of the data, columns and the encoded information in the columns.

#### 2. Exploring and Analysis Data:

- **Approach:** Conduct an initial exploration of the dataset to identify trends and relationships between bike-sharing counts and weather variables.
- **Computer operation:** Plot visualisations such as line charts, and bar chart scatter plots to understand the distribution of bike-sharing counts across different weather conditions and different times of the year.
- **Human Analyst:** Human analysts will interpret the visualisations plotted to identify potential nonlinear relationships and outliers, guiding further analysis.

#### 3. Feature Engineering:

- **Approach:** Create new features that may enhance the predictive model and polynomial regression model, such as temporal features from the 'timestamps' column.
- **Human Analyst:** A human analyst needs to understand the data to choose which features should be used to create the most accurate predictive model and features can be created from the dataset.

#### 4. Nonlinear Model:

- **Approach:** Looking into the nonlinear effects of weather variables on bike-sharing counts.
- **Computer operation:** Use polynomial regression to see nonlinear relationships. This approach will help give us insights to visualise the predicted bike-sharing counts based on temperature and humidity. The actual data points are compared to the model's predictions with the observed data.
- **Human Analyst:** Human analysts will evaluate model fit and interpret the significance of nonlinear terms, deciding on the best model based on visualisations of predicted vs. actual counts.

#### 5. Predictive Model:

- **Approach:** Develop a predictive model to forecast bike-sharing counts based on weather and temporal features.
- **Computer operation:** Implement a random forest regressor machine learning algorithm and perform hyperparameter optimisation using techniques like Grid Search (library called GridSearchCV) to enhance model accuracy.
- **Human Analyst:** Human analysts will assess model performance metrics using the RMSE performance.

### 4.2 Process

This analysis process aims to explore bike-sharing counts through three key research questions: temporal patterns, weather impact, and predictive analysis. The workflow is designed to use both computational methods and human reasoning, ensuring that insights result from data while allowing for informed decision-making throughout the analysis.

#### 1. Data Preprocessing:

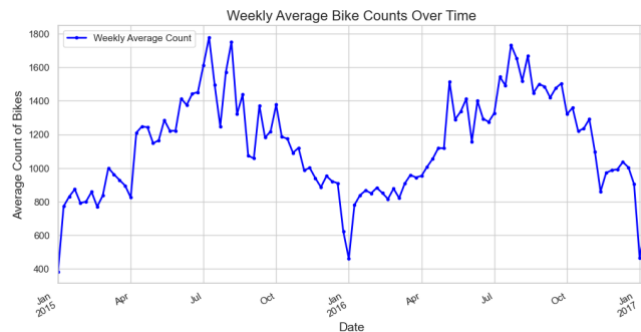
The dataset includes bike-sharing counts, weather conditions such as temperature, humidity, wind speed, and temporal features such as timestamps, seasons, and days of the week.

The dataset was cleaned to handle missing values and inconsistencies.

**Human Analyst:** Human analysts reviewed the dataset to understand the data and to ensure that the data was suitable for analysis. Pandas' library is data manipulation and gives significant information about the data. In the chosen data, it was checked for NaN values, and there were no NaN values. Also, pandas converted the column 'timestamp' to help us extract useful features such as hour, day of the week, and month, which can help capture temporal patterns in bike usage.

## 2. Exploring and Analysis Data:

Exploring and analysing data using visualisation was done to understand the dataset's structure and identify patterns. Descriptive statistics were calculated, and visualisations such as line charts to understand key metrics and distribution.



**Figure 4- Weekly Average Bike Counts Over Time**

The dataset includes a 'timestamp' field representing daily dates. If plotted directly without any preprocessing, the resulting chart would be overly crowded. To maintain the integrity and significance of the data while reducing the number of data points, we calculated a weekly average. This approach allows us to present a clearer visualisation in Figure 4. The graph shows a very recurring average count of bikes during the same time of the year but in different years.

**Human Analyst:** Human analysts interpret visualisations to identify trends, outliers, and potential relationships between variables.

## 3. Feature Engineering:

In this section, we aim to explore the nonlinear effects of weather variables, specifically temperature and humidity, on bike-sharing counts using two distinct modelling approaches: a Random Forest predictive model and polynomial regression.

By using a dataset that includes various weather conditions and bike usage statistics, we seek to find how these factors interact and influence bike-sharing demand across different seasons.

## Random Forest:

In the Random Forest model, we extract several features from the timestamp to capture the temporal dynamics of bike-sharing usage, which helps us answer our research question.

Selected the features:

- **t1:** Actual temperature.
- **t2:** "Feels like" temperature.
- **hum:** Humidity percentage.
- **wind\_speed:** Wind speed in km/h.
- **hour:** Hour of the day.
- **day\_of\_week:** Day of the week.
- **month:** Month of the year.

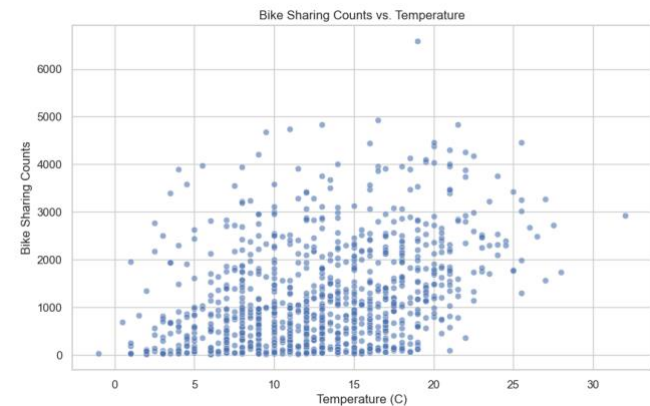
**Human Analyst:** The inclusion of temporal features allows the model to account for daily and seasonal variations in bike usage, which is important for accurate predictions.

## Polynomial Regression

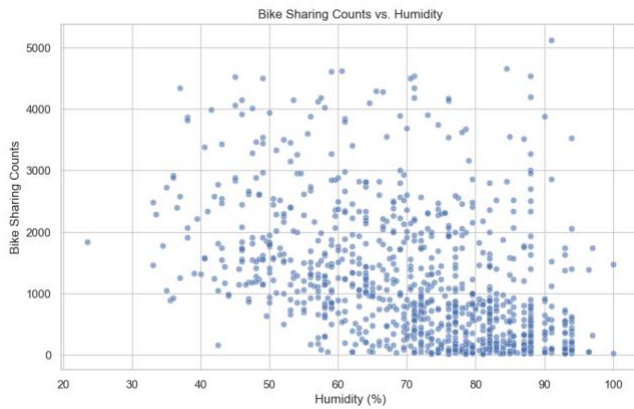
In the Polynomial Regression model, the weather variables have a nonlinear relationship with bike-sharing counts, which helps us answer our research question. To capture the nonlinear relationships, we create polynomial features of degree 2.

Selected features:

- **t1:** Actual temperature.
- **hum:** Humidity percentage.



**Figure 5- Bike Sharing Counts vs. Temperature**



**Figure 6- Bike Sharing Counts vs. Humidity**

A sample of 1000 data points has been taken from the dataset to plot both scatter plots in Figure 5 and Figure 6. Also, a random state has been placed to use the same correspondence data points to plot both figures mentioned.

**Human Analyst:** The decision to use polynomial features is based on the observed nonlinear patterns in scatter plots of temperature and humidity against bike-sharing counts as shown in Figure 5 and Figure 6. This transformation allows the model to better fit the data.

#### 4. Non-Linear Model:

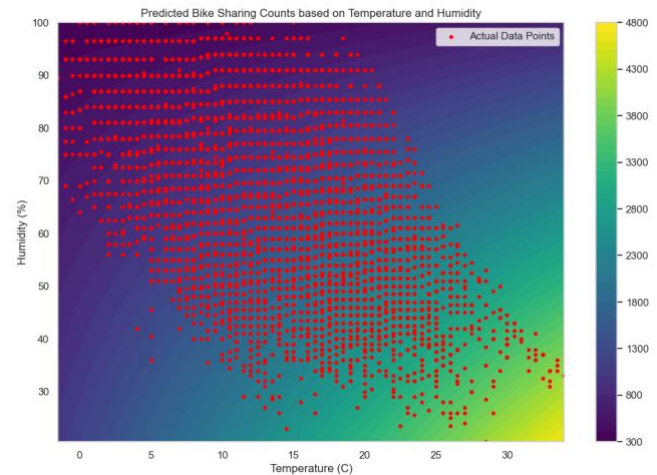
In this analysis, we aim to investigate the nonlinear effects of weather variables, specifically temperature and humidity, on bike-sharing counts using polynomial regression.

We fit the polynomial regression model using the polynomial features created earlier. After fitting the model, we analyse the coefficients and intercept to understand the influence of each feature.

**Human Analyst:** The human analyst evaluates the coefficients and intercept results yielded from the polynomial regression model.

- Intercept: 1423.28 shows if temperature and humidity were at their minimum values, the model predicts a baseline bike-sharing count of approximately 1423.
- The coefficient for Temperature ( $t_1$ ): 118.24 shows that higher temperatures are associated with increased bike usage, which is consistent with common expectations.
- The coefficient for Humidity ( $hum$ ): -11.29 shows that a negative relationship indicates that higher humidity levels are associated with lower bike usage, which aligns with the idea that uncomfortable weather conditions can discourage people from biking.
- The coefficient for Temperature ( $t_1$ ) squared: -1.17 shows that after a certain point, temperature increases may lead decrease in usage of bike-sharing counts. Specifically, as temperature increases, the effect of temperature on bike-sharing counts decreases due to this negative coefficient.

- The coefficient for Humidity ( $hum$ ) squared: 0.0038 shows that while higher humidity may correlate with increased bike usage, the impact is minimal. The positive value suggests that as humidity increases, the effect of humidity on bike-sharing counts increases slightly, but the effect is very small.



**Figure 7- Predicted Bike Sharing Counts based on Temperature and Humidity**

**Human Analyst:** The Human Analyst visualises the predicted bike-sharing counts based on temperature and humidity. Figure 7 shows the actual data points are overlaid in red, allowing us to assess how well the model's predictions align with observed values.

#### 5. Predictive Model:

With insights from the feature engineering analyses, the next step was to develop a model to predict bike-sharing counts based on weather and temporal features. A Random Forest regression model was chosen for its ability to handle non-linear relationships and interactions between variables. A train test split from the scikit learn library was used on the dataset, with 80% of the dataset on training and 20% of the dataset testing the model.

A grid search was used using the library GridSearchCV from scikit learn to choose the best hyperparameters for the model. In the Random Forest Regressor, a few hyperparameters can be used to tune for optimisation. A list was given for each hyperparameter:

- 'n\_estimators': A list of [100, 200, 300] was given which indicates the number of trees.
- 'max\_features': A list of ['log2', 'sqrt'] was given which indicates the number of features to consider at each split.
- 'max\_depth': A list of [None, 10, 20, 30] was given which indicates the maximum depth of the tree.



- 'min\_samples\_split': A list of [2, 5, 10] was given which indicates the minimum number of samples required to split an internal node
- 'min\_samples\_leaf': A list of [1, 2, 4] was given which indicates the minimum number of samples required to be at a leaf node.

A grid search is then set up and fitted into the model and after the processing of the grid search it returns the best hyperparameters uses these hyperparameters in the latest model and calculates the RMSE (Root Mean Squared Error). To further investigate a scatter plot is plotted for predicted bike-share count against actual bike-share counts with a line of best fit.

**Human Analyst:** Human analysts were involved in selecting features for the models, ensuring that the most relevant variables were included. They also set hyperparameters for the Random Forest model, such as the number of trees and maximum depth. The use of GridSearchCV allowed for systematic tuning of these parameters, but the analyst had to interpret the results to determine the best outcome.

**Stopping Condition:** The stopping condition for the modelling phase was reached when the RMSE (Root Mean Square Error) of the two models, Random Forest Regressor Predictive model and non-linear effect Polynomial Regressor on the validation set was minimised and deemed acceptable based on the context of bike-sharing counts and. The analyst compared the RMSE to baseline models and previous testing of the models to justify the model's performance.

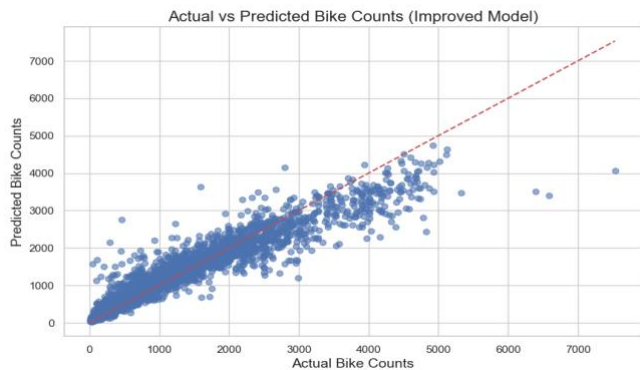
### 4.3 Results

Research question 1:

How can we predict bike-sharing counts by using weather and temporal features, and which hyperparameter optimisation methods are most effective in enhancing the accuracy of these predictions?

Building a Random Forest Regressor and training the model with these features plus additional values from the "timestamp" column, including hour, day of the week and month, allows us to predict bike-sharing counts. Grid search identified the best hyperparameters:

- 'max\_depth': None
- 'max\_features': 'log2'
- 'min\_samples\_leaf': 1,
- 'min\_samples\_split': 2,
- 'n\_estimators': 300



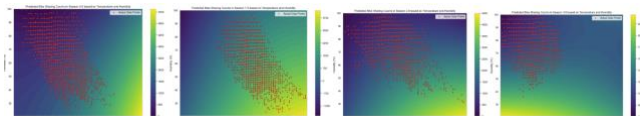
**Figure 8- Actual vs Predicted Bike Counts (Improved Model)**

With an RMSE of 348.23, the prediction model's outputs could be off by 348 counts, as shown in Figure 8. It's called "improved", as our first model showed significant improvements with an RMSE over 1000.

Research question 2:

What are the nonlinear effects of weather variables on bike-sharing counts, and how do these effects vary across different seasons and weather conditions?

The nonlinear effects are temperature and humidity as Figure 9 below shows humidity (%) on the y-axis and temperature on the x-axis; the more yellow, the more bike-share counts, with summer highest and winter lowest.



**Figure 9- Spring Summer Fall Winter Predicted Bike Sharing-Counts on Temperature and Humidity**

## 5 CRITICAL REFLECTION

This study used weather and time-based data to predict patterns and analysed the factors influencing bike-sharing use. The chosen methods, such as Random Forest modelling, polynomial regression, and visual analytics, provided useful information but indicated important weaknesses and areas that required improvement.

The models' performance was impacted by choices made about developing additional features, such as removing temporal components from the timestamp column. Polynomial regression was used once it became clear that non-linear patterns existed in the visualisations, such as scatter plots showing the relationship between temperature and bike use. Similarly, choosing the best parameter values for the Random Forest model's hyperparameter adjusting using grid search required evaluation. The key to improving the models and getting reliable outcomes was a combination of computational tools and informed choices.

In order to understand the data and conduct the analysis, visualisation was important. While scatter plots of expected against actual counts assisted in evaluating the models' accuracy, heatmaps and line charts showed seasonal trends and patterns in bike use. A more detailed review of the areas where predictions varied from observed values was achievable by these visualisations, which also made the results simpler to understand. An organised approach to the study was made sure by its ability to identify trends and patterns, which also impacted the process's later stages.

Despite the meteorological and temporal variables within the dataset were useful, the results may not have been as accurate if the locations of bike stations were provided. Also, because of overfitting, polynomial regression may not apply well to unknown data, however, it is good at capturing non-linear relationships - especially when tackling higher-degree polynomials. Furthermore, the computational costs related to grid search restricted the number of hyperparameter combinations that could be tried, hence limiting the performance of the Random Forest model.

An extensive analysis of the factors influencing bike-sharing activity would be achieved by including more data, such as real-time traffic information. A variety of parameter testing and more effectiveness could result from using different hyperparameter tuning methods, such as Bayesian optimisation. Other complex relationships can be identified by advanced methods of machine learning like deep learning. Feature engineering and visualisation creation are two examples of analysis tasks that are programmed to increase performance and minimise human labour.

To find trends and improve predictions, this research highlights the need to combine computational techniques with visual analytics. It focused on how important exploratory data analysis is and how important human judgement is in choosing the best approaches. To ensure that the results are not only correct but also useful and simple to apply, it is important to find a balance between model difficulty and the ability to interpret.

In conclusion, this study successfully examined how time and weather affect bike-sharing usage and gave valuable data. Future research could build on these results to provide even more advantages for urban mobility planning and management by tackling certain limitations and adding further data or methods.

#### Table of word counts

|                        |           |
|------------------------|-----------|
| Problem statement      | 239/250   |
| State of the art       | 476/500   |
| Properties of the data | 453/500   |
| Analysis: Approach     | 344/500   |
| Analysis: Process      | 1287/1500 |
| Analysis: Results      | 197/200   |
| Critical reflection    | 488/500   |

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